

COGNIFYZ IT SOLUTIONS

Python Development Internship

Level 2 Task Report

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Position: Python Development Intern

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Introduction

This report contains the Level 2 tasks completed during the Python Development Internship at Cognifyz IT Solutions. These tasks focused on applying Python concepts to build interactive programs involving loops, file handling, random number generation, password validation, and sequence generation.

Task 1: Guessing Game

Objective

Create a game where the computer generates a random number and the user attempts to guess it.

Description

The program generates a random number between 1 and 100 and provides hints such as "Too High" or "Too Low" until the correct number is guessed.

Concepts Used

- Random Module
 - Loops
 - Conditional Statements
 - User Input
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Task 2: Number Guesser

Objective

Develop a number guessing game using a user-defined range.

Description

The program generates a random number within a specified range entered by the user. The user then attempts to guess the number while receiving appropriate feedback.

Concepts Used

- Random Module
 - Loops
 - Conditional Statements
 - User Input
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Task 3: Password Strength Checker

Objective

Evaluate the strength of a password based on predefined criteria.

Description

The program checks whether a password contains uppercase letters, lowercase letters, digits, special characters, and meets the minimum length requirement.

Concepts Used

- Strings
 - Loops
 - Conditional Statements
 - User Input
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Task 4: Fibonacci Sequence

Objective

Generate the Fibonacci sequence up to a specified number of terms.

Description

The program accepts the number of terms from the user and displays the Fibonacci sequence accordingly.

Concepts Used

- Loops
 - Variables
 - Arithmetic Operations
 - User Input
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Task 5: File Manipulation

Objective

Read a text file and count the occurrences of each word.

Description

The program reads a text file, counts how many times each word appears, and displays the results in alphabetical order.

Concepts Used

- File Handling
 - Dictionaries
 - Loops
 - String Manipulation
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Conclusion

The completion of Level 2 tasks provided practical experience with Python programming concepts such as file handling, password validation, random number generation, looping structures, and data processing. These tasks helped strengthen problem-solving skills and improve understanding of real-world programming applications.